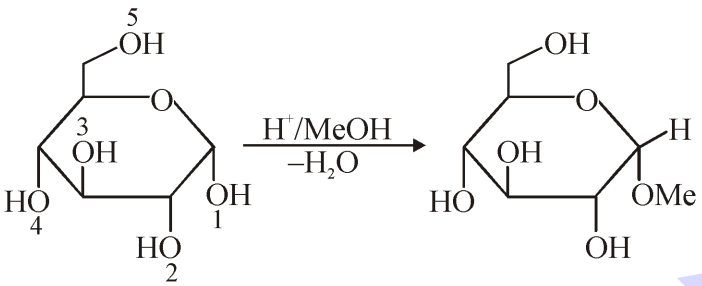
SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. (D)

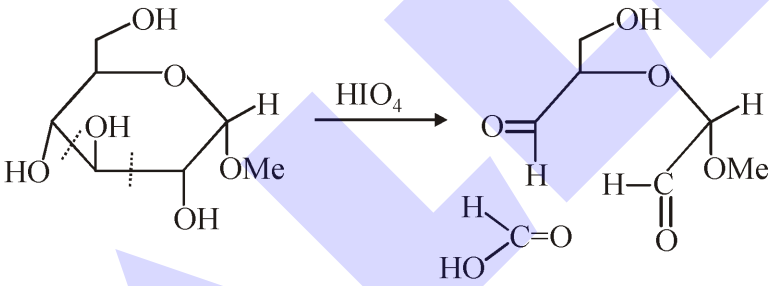
Sol.



Hemiacetal

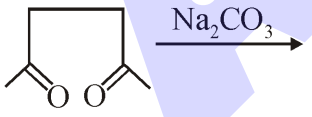
Acetal (will not show Tollen's Test)

$-\overset{1}{OH}$ Will be methylated because of formation of most stable carbocation. Intermediate is carbocation.



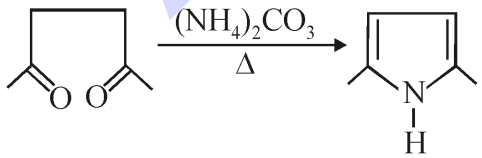
2. (C) II, III, IV, V

Sol. (I)

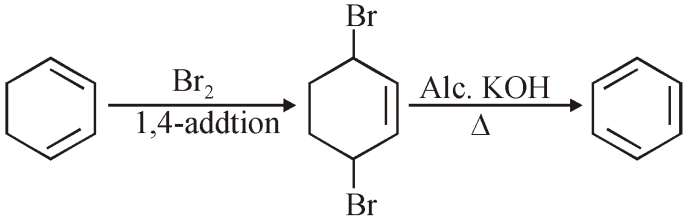


Intra molecular aldol

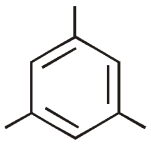
(II)

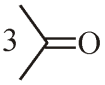
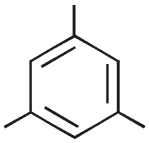


(III)



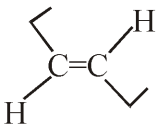
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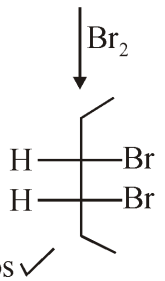
(IV) $CH_3–C\equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$ 

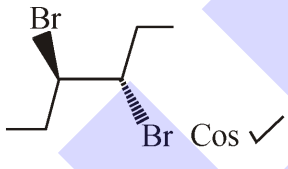
(V) 3  $\xrightarrow{\text{Conc. } H_2SO_4}$ 

3. (B)

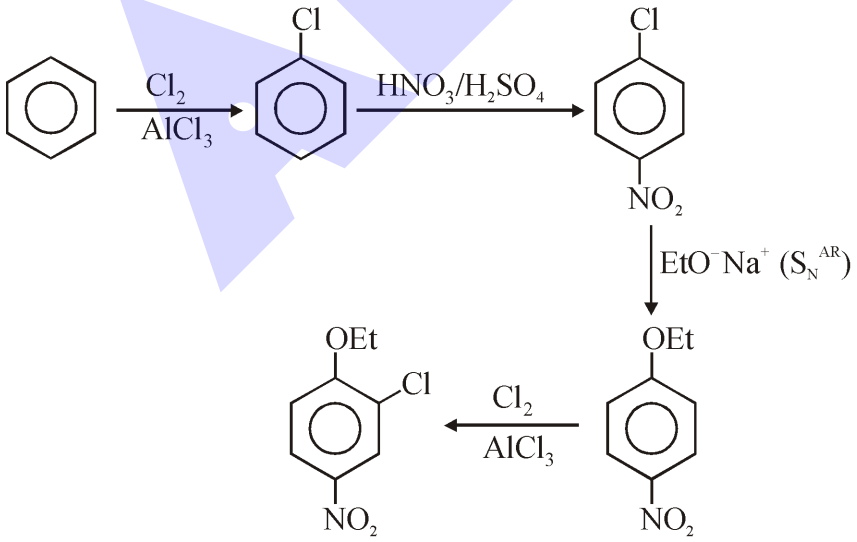
Sol. $CH_3–CH_2–C\equiv CH \xrightarrow{NaNH_2} CH_3–CH_2–C\equiv C^{\ominus}$

$CH_3–CH_2–C\equiv C^{\ominus} \xrightarrow{Et–I} Et–C\equiv C-Et \xrightarrow[\text{Liq. } NH_3]{Na}$ 



Pos ✓ Meso =  Br Cos ✓ Meso

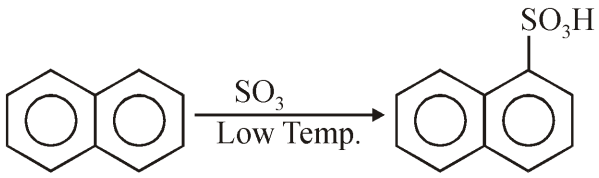
4. (C)

Sol. 

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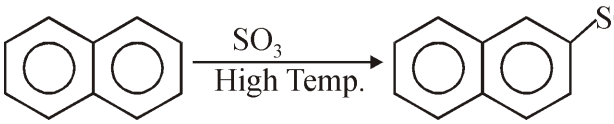
5. (A)

Sol.



is Kcp

(More stable σ-complex carbocation)

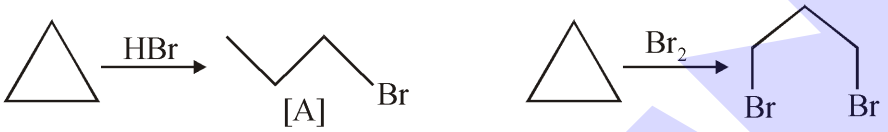


is Tcp

(More stable product)

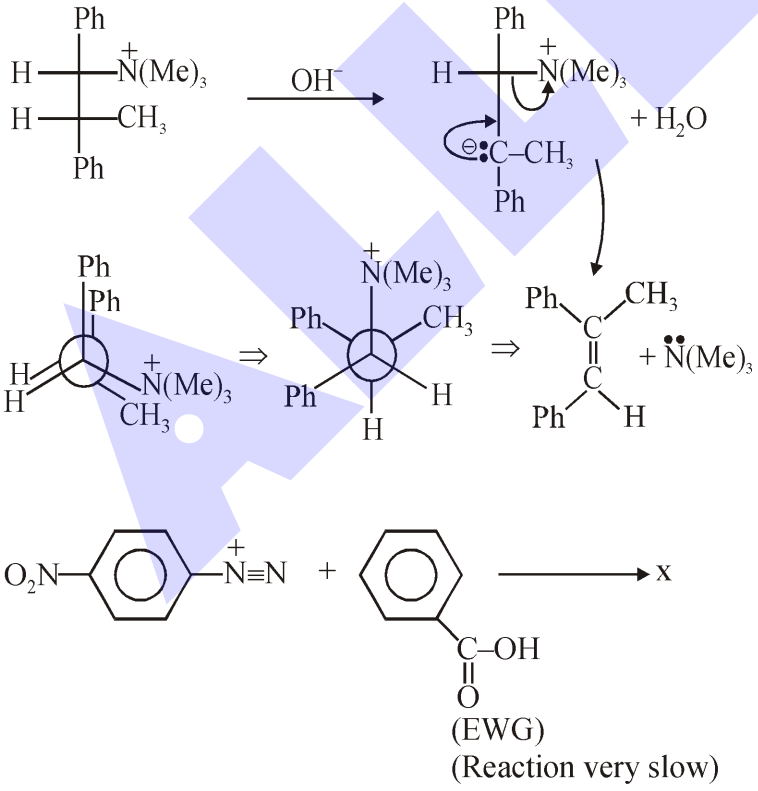
6. (B,C)

Sol.



7. (A,B,C)

Sol. (D)

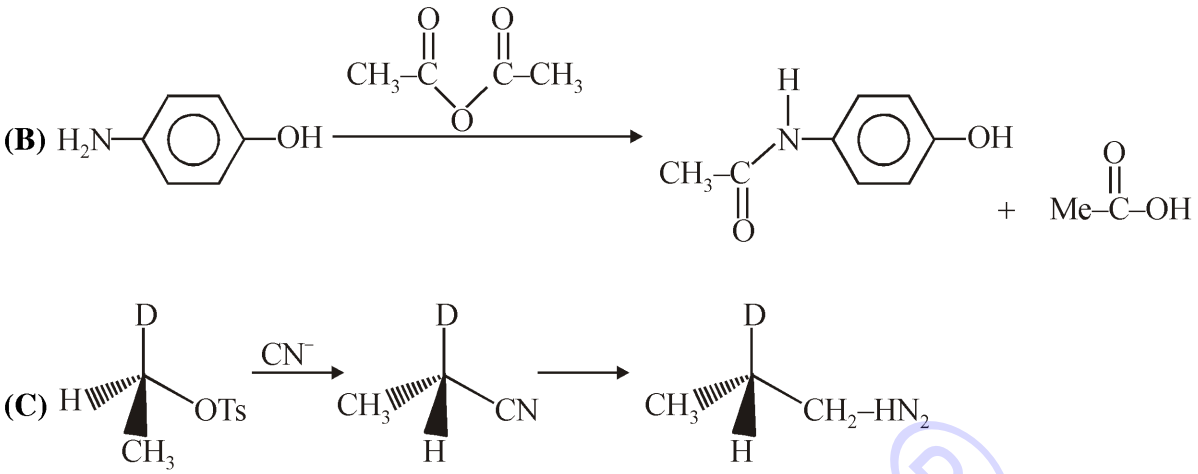


(A)

(EWG)

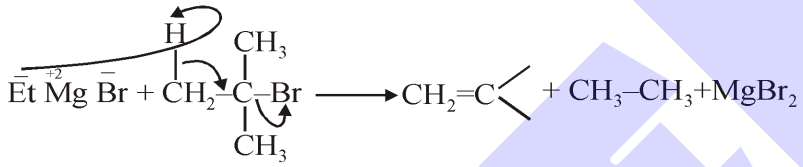
(Reaction very slow)

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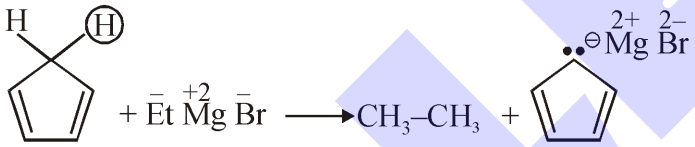


8. (A,B,D)

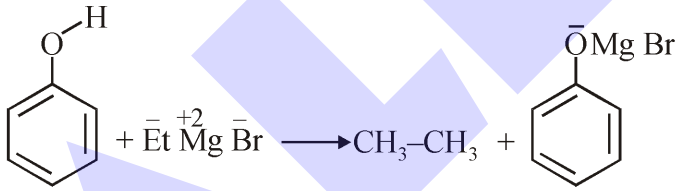
Sol. (A)



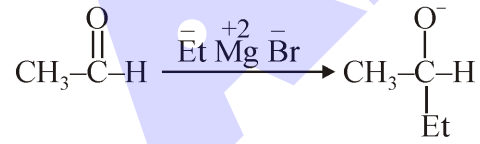
(B)



(D)

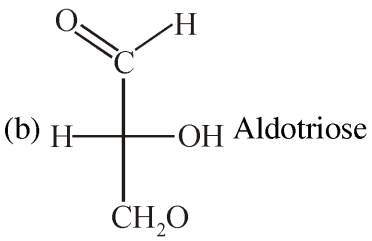


(C)



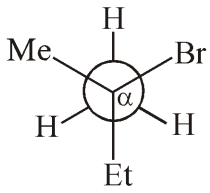
9. **(A,B,C)**

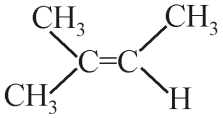
Sol. (a) Deoxyribose is present in DNA



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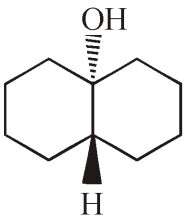
(c) Acid strength  >   Aromatic nature

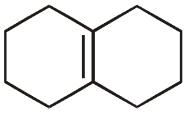
(D) 

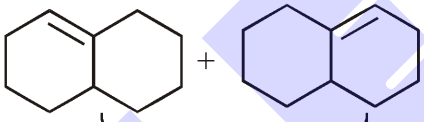
Alc. KOH, Δ, E^2 → 

NO. GI

10. (A,B,D)

Sol. 

Conc. H_3PO_4 →  Major Saytzeff's

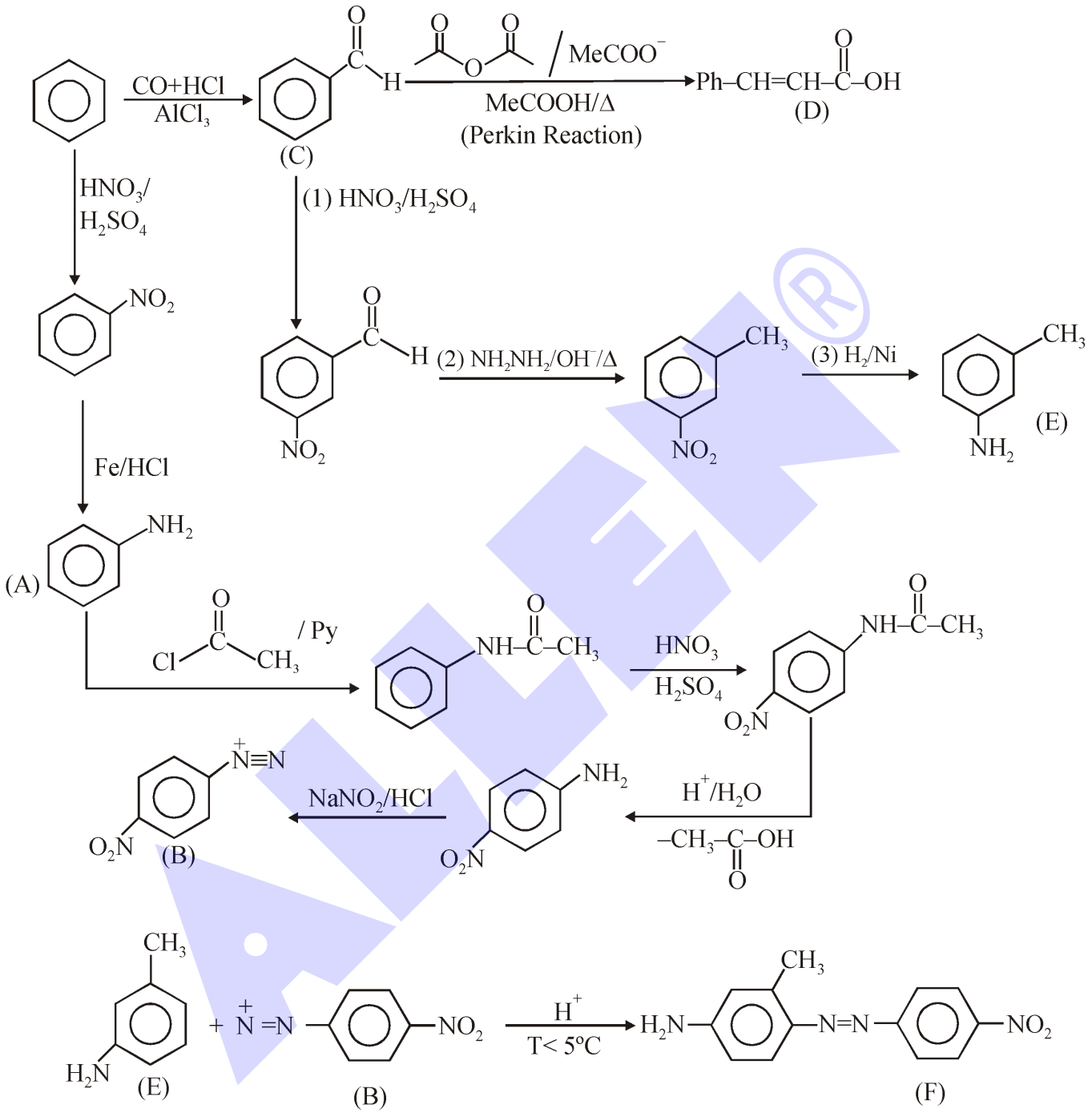
 +

Enantiomers
Hoffmann

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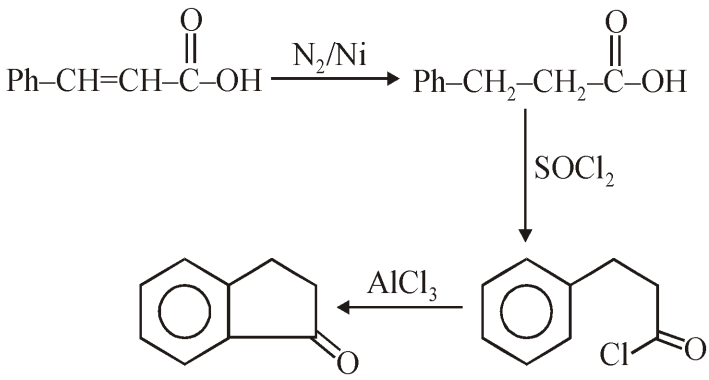
11. (A)

Sol.

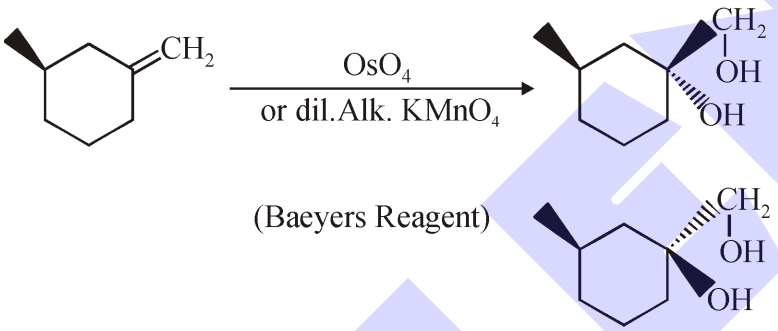


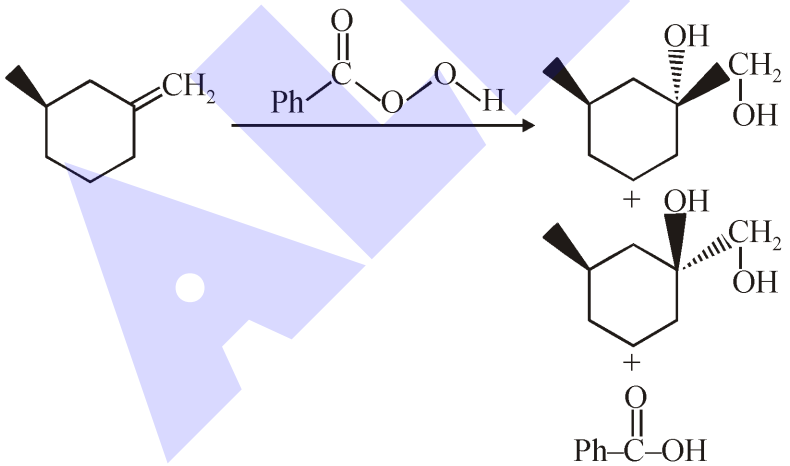
For More Material Join: @JEEAdvanced_2025

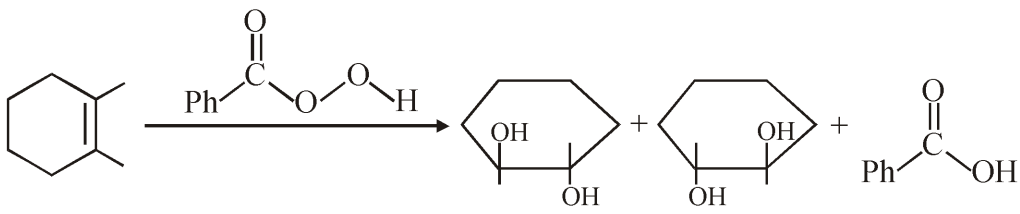
12. (C)

Sol. 

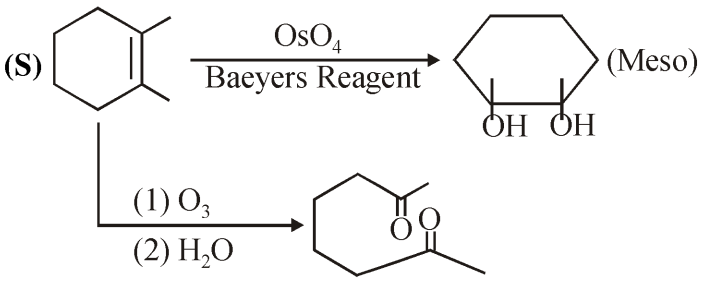
13. (B)

Sol. (P) 

(Q) 

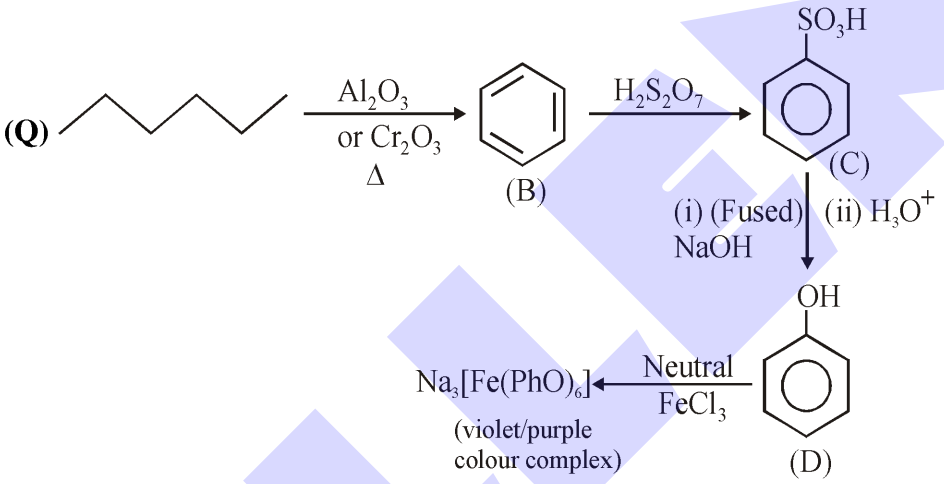
(R) 

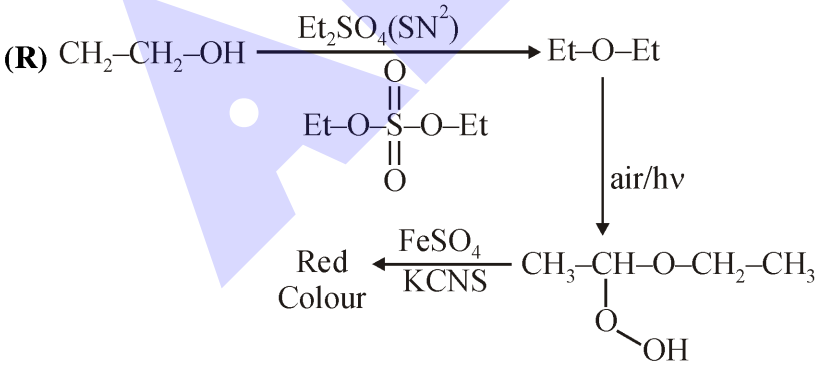
For More Material Join: @JEEAdvanced_2025

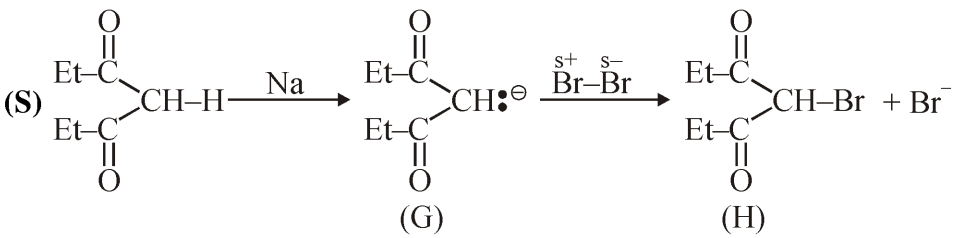


14. (B)

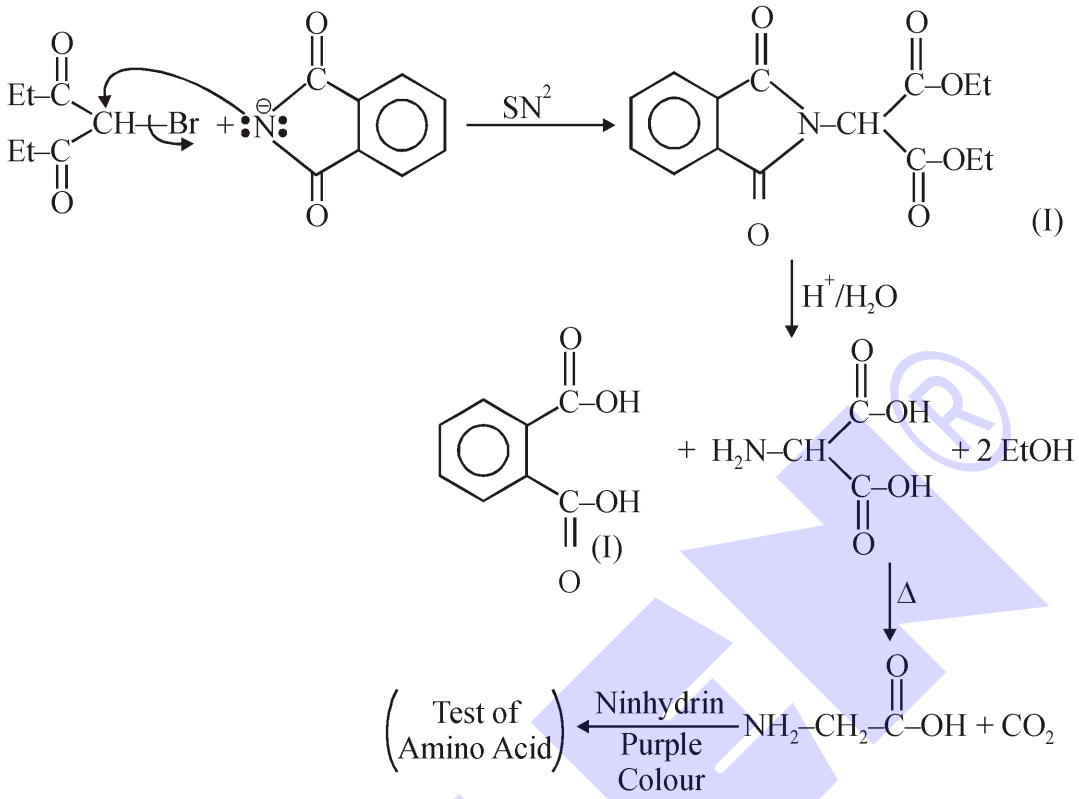
Sol. **(P)** $Mg_2C_3 + 2H_2O \rightarrow CH_3–C\equiv CH + 2Mg(OH)_2$







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Section-III

Integer value correct type

1. (6)

Sol. All correct

2. **(5)**

Sol. (i), (iii),(iv),(v),(vi)

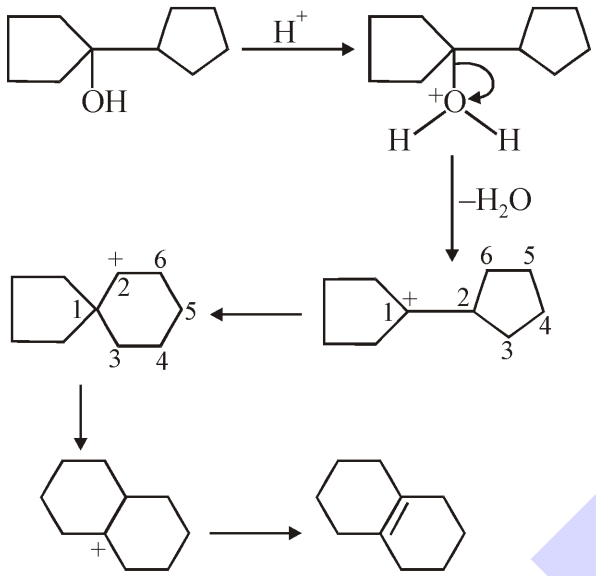
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3. (5)

Sol. (i), (ii), (iii),(v),(vi)

4.

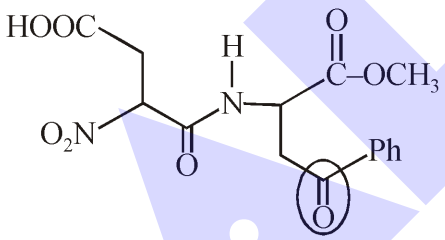
Sol.



Number of steps = 5 ; number of Ts = 5

5.

Sol.



Only ketone is reduced by $NaBH_4$

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